Journal of Psychiatric Research 195 (2026) 74–84

Contents lists available at ScienceDirect

Journal of Psychiatric Research

journal homepage: www.elsevier.com/locate/jpsychires

A systematic review of transcranial direct current stimulation (tDCS) for adults with attention deficit/hyperactivity disorder (ADHD)

Tianran Zhang [a], Dimitrios Adamis [b,*]

[a] Department of Health & Nutritional Sciences, Atlantic Technological University, Sligo, F91 YW50, Ireland
[b] HSE Sligo Mental Health Services, University of Galway, University of Limerick, University College Dublin, Ireland

ARTICLE INFO

Keywords:
ADHD
Transcranial direct current stimulation
Systematic review
cognitive enhancement
Neuromodulation

ABSTRACT

Background: Transcranial direct current stimulation (tDCS) has been proposed as a potential non-pharmacological intervention for Attention-Deficit/Hyperactivity Disorder (ADHD). While evidence from pediatric populations is growing, the effects of tDCS in adults with ADHD remain insufficiently characterized, increasing recognition of ADHD as a lifelong neurodevelopmental condition with persistent functional impairments.

Objective: This systematic review aimed to evaluate the effects of tDCS in adults with ADHD across clinical, cognitive, and neurophysiological outcome domains.

Methods: Six electronic databases were systematically searched to identify peer-reviewed controlled trials evaluating tDCS interventions in adults with clinically diagnosed ADHD through May 2025. Data extraction encompassed clinical symptom outcomes, cognitive task performance, neurophysiological findings, adverse events, medication status, and stimulation protocols. Study quality was assessed using the Downs and Black checklist. Where sufficient data were available, an exploratory random-effects meta-analysis was conducted for clinical symptom outcomes; all other outcomes were synthesized narratively.

Results: Eleven controlled studies (n = 415) met inclusion criteria and were included in the qualitative synthesis. Two randomized controlled trials provided sufficient data for exploratory meta-analysis of clinical symptoms. Pooled results suggested a possible beneficial effect of tDCS on inattention symptoms, accompanied by substantial heterogeneity (I^2 = 79.95 %), while no significant effect was observed for hyperactivity/impulsivity. Cognitive outcomes were heterogeneous and domain-specific, with some evidence of post-stimulation improvements in inhibitory control, working memory, and attentional measures. Two studies reported tDCS-induced modulation of frontal neurophysiological markers associated with cognitive control. Stimulation protocols varied considerably across studies, with no consistent association between specific parameters and outcomes. Adverse events were generally mild and transient, and follow-up data were limited.

Conclusions: Current evidence suggests that tDCS may exert domain-specific effects on inattention and selected cognitive functions in adults with ADHD. However, conclusions remain tentative due to methodological heterogeneity, small sample sizes, and limited quantitative evidence. Future research should prioritize standardized protocols, adequately powered trials with extended follow-up periods, integration of neurophysiological biomarkers, and systematic evaluation of individual difference factors to clarify the therapeutic potential and mechanisms, and advance precision neuromodulation approaches for adult ADHD.

1. Introduction

Attention-deficit/hyperactivity disorder (ADHD) is a neurodevelopmental condition characterized by persistent patterns of inattention, and/or hyperactivity, and impulsivity that significantly impair functioning (American Psychiatric Association, 2013). Once considered primarily a childhood disorder, accumulating evidence demonstrates its persistence into adolescence and adulthood, often resulting in substantial academic, occupational, and social difficulties (Sibley et al., 2017; Turgay et al., 2012).

Neuroimaging and electrophysiological studies have consistently reported abnormalities in frontostriatal and frontoparietal networks in

* Corresponding author.
E-mail address: dimaadamis@yahoo.com (D. Adamis).

https://doi.org/10.1016/j.jpsychires.2026.01.030
Received 28 October 2025; Received in revised form 21 January 2026; Accepted 22 January 2026
Available online 23 January 2026
0022-3956/© 2026 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

Downloaded for Anonymous User (n/a) at Stony Brook University from ClinicalKey.com by Elsevier on October 22, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

individuals with ADHD, including altered activity in the dorsolateral prefrontal cortex (DLPFC), the anterior cingulate cortex, and the basal ganglia, regions implicated in executive function, attention regulation, and impulse control (Marcos-Vidal et al., 2018; Passarotti et al., 2009; Seidman et al., 2005; Samea et al., 2019; Rapport et al., 2013; Kasper et al., 2012; Keune et al., 2015; Krain and Castellanos, 2006). These cognitive domains also represent core features of ADHD pathophysiology and are central to most theoretical models of the disorder (Barkley, 1997; Willcutt et al., 2005; Rubia, 2018). These neural features offer a neurobiological rationale for targeting cortical regions through brain-based interventions.

Current ADHD treatment strategies primarily comprise pharmacological interventions, including stimulant and non-stimulant medications, and psychological therapies, particularly cognitive-behavioural approaches (Sibley et al., 2014; Fullen et al., 2020). While pharmacotherapy demonstrates robust short-term efficacy, adverse effects, non-adherence, potential misuse, and contraindications limit its utility in certain populations (Hsu et al., 2017). Behavioural interventions, though effective for skill-building and behavior management, often require sustained therapeutic engagement, trained personnel, and structured environments, which may be less accessible to adults or insufficient for addressing core neurocognitive deficits (Daley et al., 2014). These limitations have sparked interest in alternative and adjunctive interventions that can directly modulate underlying neural dysfunction.

Non-invasive brain stimulation techniques, particularly transcranial direct current stimulation (tDCS), have emerged as promising neuromodulatory treatments for ADHD (Westwood et al., 2021; Salehinejad et al., 2019, 2020). tDCS modulates cortical excitability through weak direct current applied via scalp electrodes, with polarity-dependent effects such that anodal stimulation is generally associated with increased neuronal excitability, while cathodal stimulation is associated with relative reductions in excitability at the cortical level. Although these effects are context-dependent and influenced by baseline neural state, stimulation parameters, and task engagement, this basic neurophysiological framework has guided the design of most tDCS protocols targeting prefrontal regions. Its portability, favourable safety profile, and low cost make it an attractive candidate for clinical implementation (Polanía et al., 2018; Imburgio and Orr, 2018; Bikson et al., 2016). Emerging evidence suggests that tDCS may enhance cognitive performance and reduce core symptoms in individuals with ADHD by influencing prefrontal networks involved in attention and executive control (Salehinejad et al., 2019; Rubia, 2018; Nejati et al., 2017; Breitling et al., 2016), with a substantial proportion of existing research focusing on pediatric or mixed-age populations.

Although research on tDCS in children and adolescents with ADHD has expanded, evidence in adult populations remains comparatively limited (Courrèges et al., 2025). This imbalance is notable given the growing recognition of ADHD as a chronic condition with persistent cognitive and functional impairments across the lifespan (Wender et al., 2001). While core symptom domains are shared across age groups, developmental changes in brain organization and cognitive control systems suggest that findings derived from pediatric samples may not be directly generalizable to adults. These considerations underscore the need for focused synthesis of the adult literature, rather than reliance on extrapolation from child and adolescent studies.

Adults with ADHD face elevated risks of educational underachievement, occupational instability, interpersonal difficulties, and comorbid psychiatric disorders (Kooij et al., 2010). Epidemiological data indicate an average adult ADHD prevalence of 2.8 % across 10 countries in the WHO World Mental Health Surveys (Fayyad et al., 2016), while meta-analytic evidence reveals prevalence rates of 27 % in psychiatric outpatient populations when screening tools are used, decreasing to 14.61 % with formal diagnostic assessment (Adamis et al., 2022). Despite these substantial figures, adult ADHD remains significantly underdiagnosed and undertreated, creating considerable socioeconomic burden and unmet clinical need.

Within this context, tDCS has been investigated as a potential yet underexplored intervention for adult ADHD. However, the more modest and heterogeneous effects reported in adult studies, relative to pediatric samples, raise questions about age-related factors that may influence responsiveness to neuromodulation. One plausible consideration is that adults typically exhibit reduced experience-dependent cortical plasticity and greater reliance on established functional networks compared with children, which may constrain the magnitude or durability of stimulation-induced effects. Importantly, this interpretation remains tentative, as the adult tDCS literature is characterized by substantial variability in stimulation protocols, target brain regions, outcome measures, and participant characteristics. Furthermore, how stimulation parameters (site, polarity, intensity, duration) and individual factors (medication status, symptom subtype) influence treatment outcomes is not yet well characterized (Salehinejad et al., 2020).

To address these gaps, we conducted a systematic review of controlled studies investigating tDCS effects in adult ADHD. In contrast to broader neuromodulation reviews and syntheses that include mixed-age populations or focus on narrower outcome domains, the present review focuses specifically on adults and considers clinical, cognitive, and neurophysiological outcomes alongside medication status and stimulation parameters. Our objectives were to: (1) examine whether tDCS improves cognitive functions, neural activity, or clinical symptoms in adults with ADHD; (2) identify the cognitive domains and neurophysiological measures most responsive to stimulation; and (3) explore how stimulation parameters and participant characteristics modulate treatment effects. This synthesis aims to clarify current findings and identify priorities for future research and clinical applications.

2. Methods

2.1. Search strategy

This systematic review adhered to PRISMA 2020 guidelines (Page et al., 2021) and was pre-registered on PROSPERO (2025 CRD420251070474; available from https://www.crd.york.ac.uk/PROSPERO/view/CRD420251070474). We conducted comprehensive searches across six electronic databases: PubMed, MEDLINE (via Ovid), EBSCO (including PsycINFO), Web of Science, Scopus, and Cochrane CENTRAL, covering all records through May 2025. The search strategy combined terms related to “ADHD,” “attention deficit hyperactivity disorder,” “transcranial direct current stimulation,” “tDCS,” and “adult,” utilizing both keywords and database-specific subject headings. Full search strategies for each database are provided in Supplementary Material (Search Strategy).

2.2. Selection process

Following duplicate removal based on titles and publication details, records underwent manual screening in three sequential phases: title review, abstract screening, and full-text assessment. Two independent reviewers applied predefined eligibility criteria, with discrepancies resolved through a consensus process.

Inclusion criteria: Studies were eligible if they (1) involved adults aged 18 years and older with formal ADHD diagnoses using established criteria (DSM or ICD), (2) applied tDCS as a primary or standalone intervention, (3) employed a control or comparison group (sham stimulation, waitlist, or baseline condition), (4) reported quantitative outcomes for clinical symptoms (inattention, impulsivity, hyperactivity) or cognitive performance (e.g., executive functioning, working memory, attention), and (5) were peer-reviewed primary studies.

Exclusion criteria: We excluded (1) studies conducted exclusively with children or adolescents, (2) animal studies, (3) case studies, reviews, conference abstracts without full text, or editorials, (4) studies where tDCS effects could not be isolated (e.g., combined interventions

Downloaded for Anonymous User (n/a) at Stony Brook University from ClinicalKey.com by Elsevier on October 22, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

precluding comparison), and (5) non-English publications.

2.3. Data extraction and statistical analysis

Data extraction employed a structured template that captured the study design, participant characteristics, diagnostic criteria, stimulation parameters (including montage, intensity, duration, and session number), control conditions, primary and secondary outcomes, adverse events, and medication status. Medication status was categorized as unmedicated, medicated-withdrawn, stable medication, medication-naïve, or partially medicated. Additional variables included ADHD subtype and exclusion criteria for comorbidities. Two reports by Cosmo et al. (2015a, 2015b), although derived from the same participant sample, were retained as separate studies because they reported non-overlapping outcome domains (cognitive task performance vs. EEG measures), and therefore did not result in duplication within the quantitative synthesis.

Statistical analyses were performed using the "meta" package in R software (version 4.5.1). For clinical symptoms, we conducted an exploratory random-effects meta-analysis using post-intervention symptom scores from two eligible trials, which employed closely matched ADHD rating scales (ASRS and CASRS) with comparable item structure and scoring direction. Standardized mean differences (Hedges' g) with 95 % confidence intervals were calculated, and heterogeneity was assessed using the I^2 statistic. Given the limited number of eligible studies and minor differences in rating source (self-report vs. clinician-rated), the meta-analytic results were interpreted cautiously, with remaining clinical findings synthesized narratively.

Cognitive outcomes underwent structured narrative synthesis. First, we extracted and coded all reported tasks per study based on statistical significance at the post-intervention time point ($p < .05$), prioritizing between-group comparisons when available. Second, tasks were categorized into five predefined cognitive domains: inhibitory control, working memory, attention, executive function, and social/emotional cognition. For each domain, we summarized the number of tasks employed, proportion showing significant improvement, and available effect sizes. When multiple dependent measures were derived from the same task (e.g., multiple Tower of London indices), these were treated as a single task to maintain consistency in domain-level aggregation.

Neurophysiological outcomes from two studies using electroencephalography (EEG) were synthesized narratively. Neural correlates included spectral power (delta, theta, beta bands), resting-state EEG changes, and event-related potential (ERP) components (P200, P300) during cognitive or emotional tasks. Methodological heterogeneity precluded pooled analysis.

Additional narrative synthesis examined stimulation protocols, adverse events, and potential effect moderators, including medication status, participant demographics, and ADHD subtypes. These elements were analysed descriptively to assess the consistency of tDCS effects across methodological variations. Given design heterogeneity and variable reporting standards, all subgroup and moderator analyses remained exploratory.

2.4. Risk of bias assessment

Risk of bias and methodological quality were assessed using the Downs and Black checklist (Downs and Black, 1998), which allows structured appraisal across reporting quality, external validity, internal validity (bias and confounding), and statistical power. All included studies were assessed independently by two reviewers, with discrepancies resolved through discussion to achieve consensus. Item-level scores for each study are provided in Supplementary Table S1. Across the eleven included studies, total Downs and Black scores ranged from 17 to 28 (out of a possible 32), indicating overall moderate to high methodological quality. Reporting quality was generally strong, with most studies scoring between 9 and 10 out of 10, reflecting clear descriptions of objectives, interventions, outcomes, and statistical analyses. External validity scores were consistently moderate to high (2–3 out of 3), suggesting reasonable generalizability within clinical and research settings.

Internal validity related to bias was also rated highly across studies (typically 7 out of 7), indicating appropriate randomization procedures, outcome assessment, and control of measurement bias. Similarly, internal validity concerning confounding was generally strong (5–6 out of 6), reflecting adequate control of key confounders and appropriate analytic strategies in most trials.

In contrast, statistical power emerged as the most consistently limited domain, with power scores ranging from 0 to 2, and the majority of studies scoring 0 or 1. This reflects the absence of formal power calculations and the small sample sizes characteristic of many included trials, particularly those assessing clinical symptom outcomes. Lower overall scores were therefore primarily driven by limitations in power rather than deficiencies in study design or reporting.

Taken together, this quality assessment suggests that while the included studies were generally well conducted and clearly reported, limitations related to sample size and statistical power constrain the strength and precision of effect estimates. These methodological features are considered further in the Discussion when interpreting the robustness and clinical implications of the synthesized evidence.

3. Results

Database searches identified 1176 records. Following screening and full-text assessment, 11 studies met inclusion criteria and were included in the final review (Leffa et al., 2022; Cachoeira et al., 2017; Basiri and Hadianfard, 2023; Allenby et al., 2018; Barham et al., 2022; Beiman et al., 2025; Cosmo et al., 2015; Cosmo et al., 2015b; Dubreuil-Vall et al., 2021; Weinberg et al., 2023; Jacoby and Lavidor, 2018). The study selection process is summarized in the PRISMA flow diagram (Fig. 1).

3.1. Study characteristics

Among the 11 included studies, 3 assessed clinical symptoms as primary outcomes, 9 evaluated various cognitive functions, and 2 included EEG-based neurophysiological measurements. Across studies, tDCS protocols varied substantially, with stimulation intensities ranging from 1 to 2 mA, session durations from 20 to 30 min, and the number of sessions ranging from single-session designs to multi-session interventions (up to 28 sessions), reflecting considerable heterogeneity in intervention parameters. Study details are presented in Table 1.

Medication status categories: Unmedicated: Participants not taking medication during study period or discontinued for ≥1 month (sufficient washout). Stable medication: Participants remained unmedicated or maintained stable regimen for defined duration (≥4 weeks). Medication-naïve: Participants never received prior ADHD pharmacological treatment. Medicated (withdrawn): Participants unmedicated or withheld medication on experimental days (24-h discontinuation). Partially medicated: Mixed sample with no medication status control or standardization.

3.2. Exploratory meta-analysis of clinical outcomes

We conducted an exploratory random-effects meta-analysis of clinical symptom outcomes using post-intervention scores from two randomized controlled trials (RCT) (Leffa et al., 2022; Cachoeira et al., 2017) across two domains: inattention and hyperactivity/impulsivity. Given the limited number of eligible studies reporting sufficiently comparable clinical outcomes, this analysis was undertaken to provide a quantitative summary of available data rather than a definitive estimate of treatment effects.

For inattention symptoms, the pooled effect estimate favored active tDCS over sham (Hedges' $g = -1.62$, 95 % CI [–3.13, –.10], $p = .037$);

Downloaded for Anonymous User (n/a) at Stony Brook University from ClinicalKey.com by Elsevier on October 22, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

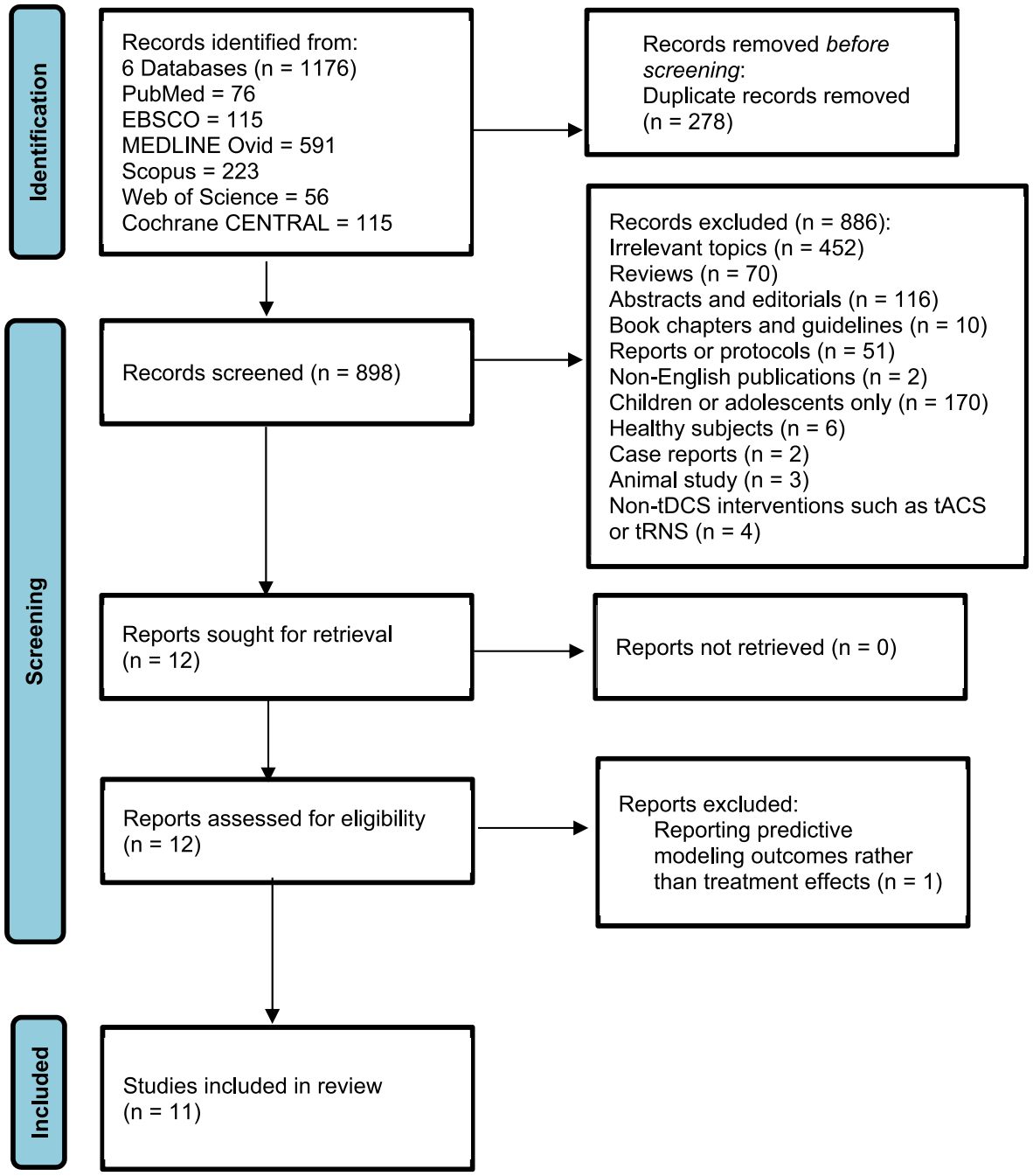


Fig. 1. PRISMA 2020 Flow diagram.

however, the confidence interval was wide and substantial heterogeneity was observed ($I^2 = 79.95$ %), indicating considerable imprecision and sensitivity of the estimate to individual studies. For hyperactivity/impulsivity symptoms, the pooled effect estimate was smaller and non-significant (Hedges' g = −.94, 95 % CI [–2.41, .53], $p = .211$), with similarly high heterogeneity ($I^2 = 82.84$ %). Effect estimates are visualized in Figs. 2 and 3.

These pooled estimates are directionally consistent with findings reported in the original study. Leffa et al. reported a large post-intervention effect for inattention ($d = 1.23$) and a small, non-significant effect for hyperactivity/impulsivity ($d = .33$); while Cachoeira et al. reported moderate effects for both domains ($d = .85$ and .60, respectively), though exact p-values were unavailable. Importantly, the pooled estimates do not provide additional precision beyond the individual study results and should be interpreted as a descriptive quantitative summary.

The substantial heterogeneity likely reflects methodological and clinical differences between the two trials. Although both studies targeted the DLPFC using F3/F4 montages, Leffa et al. employed a home-based intervention (28 sessions over 4 weeks, 30 min daily), while Cachoeira et al. used a laboratory-based protocol (5 consecutive days, 20 min daily). Additionally, outcome measures differed between clinician-rated (CASRS) and self-report (ASRS) scales, which may vary in sensitivity to change due to differences in rater perspective and contextual anchoring of symptom severity. Cachoeira et al. also found that clinical benefits declined over 4-week follow-up, suggesting sustained stimulation may be necessary for lasting improvement.

In addition to the two studies included in the meta-analysis, Basiri et al. (2023) evaluated clinical outcomes using ASRS but could not be quantitatively synthesized due to insufficient statistical reporting. This

Downloaded for Anonymous User (n/a) at Stony Brook University from ClinicalKey.com by Elsevier on October 22, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

Table 1
Study Characteristics of RCTs Investigating tDCS in Adults with ADHD.

Authors (Year)	Country	Study Design	Sample Characteristics			Clinical Features			Assessment Tools			Intervention Parameters			
			N Total (ADHD)	Mean (SD) Age	Sex (% male)	Comorbidities	Medication Status	Diagnostic Criteria for Screen	Clinical Symptom Measures	Cognitive Measures	Neurophysiological Measures (EEG)	Stimulation Site	Intensity (mA)	Duration (min)	Number of Sessions
Leffa et al. (2022)	Brazil	Parallel-group RCT	64	38.3 (9.6)	53 %	No	Unmedicated	DSM-5	CASRS	NA	NA	F4	2	30	28
Tosetto Cachoeira et al. (2017)	Brazil	Parallel-group RCT	17	32.29 (4.99)	46.82 %	No	Unmedicated	DSM-5	ASRS	NA	NA	F4	2	20	5
Basiri and Hadianfard. (2023)	Iran	Parallel-group RCT	80	23.5 (2.9)	21.25 %	No	Unmedicated	DSM-5	ASRS	ST, CPT	NA	F3	1.5	20	10
Allenby et al. (2018)	USA	Within-subject RCT	37	31.7	70 %	No	Stable medication	DSM-5	NA	Conners' CPT, SST	NA	F3	2	20	3
Barham et al. (2022)	Turkey	Parallel-group RCT	22	22 (2.77)	32 %	No	Medication-naïve	DSM-5	NA	DSF, TMT-A, ST, TMT-B, DSB, CPT, ToL, RMET	NA	F4	2	20	5
Beiman et al. (2025)	Israel	Parallel-group RCT	25	23.16 (2.27)	28 %	No	Medicated (withdrawn)	DSM-5	NA	CPT, N-back, SST, ST	NA	F3	2	20	1
Cosmo et al. (2015)	Brazil	Parallel-group RCT	60	32.25 (10.88)	58.33 %	No	Partially medicated	DSM-IV	NA	GNG	NA	F3	1	20	1
Cosmo et al. (2015)	Brazil	Parallel-group RCT	60	32.2 (10.9)	58.33 %	No	NR	DSM-IV	NA	NA	Functional Cortical Network Model	F3	1	20	1
Dubreuil-Vall et al. (2021)	USA	Within-subject RCT	44	37.5 (15.16)	50 %	No	Medicated (withdrawn)	DSM-5	NA	EFT/SST	ERP	F3/F4	2	30	1
Weinberg et al. (2023)	Israel	Parallel-group RCT	22	NR	NR	No	Medicated (withdrawn)	NR	NA	SST, Control task	NA	Crossing point between F4-F8	1	10	1
Jacoby and Lavidor (2018)	Israel	Within-subject RCT	20	22.75 (2.8)	55 %	No	Medicated (withdrawn)	NR	NA	MOXO-CPT	NA	Bilateral DLPFC	1.8	20	1

Note. NR = Not Reported; NA = Not Assessed; CASRS = Clinical-administered Adult ADHD Self-report Scale; ASRS = Adult ADHD Self-report Scale; CPT = Continuous Performance Task; SST = Stop Signal Task; ST = Stroop Test; DSF = Digit Span – Forward; DSB = Digit Span – Backward; TMT-A = Trail Making Test – Form A; TMT-B = Trail Making Test – Form B; N-back = N-back Task; GNG = Go/No-Go Task; ToL = Tower of London Test; RMET = Reading the Mind in the Eyes Test; EFT = Eriksen Flanker Task; ERP = Event-Related Potentials; DLPFC = Dorsolateral Prefrontal Cortex. The stimulation site refers to 10–20 EEG system coordinates.

Downloaded for Anonymous User (n/a) at Stony Brook University from ClinicalKey.com by Elsevier on October 22, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

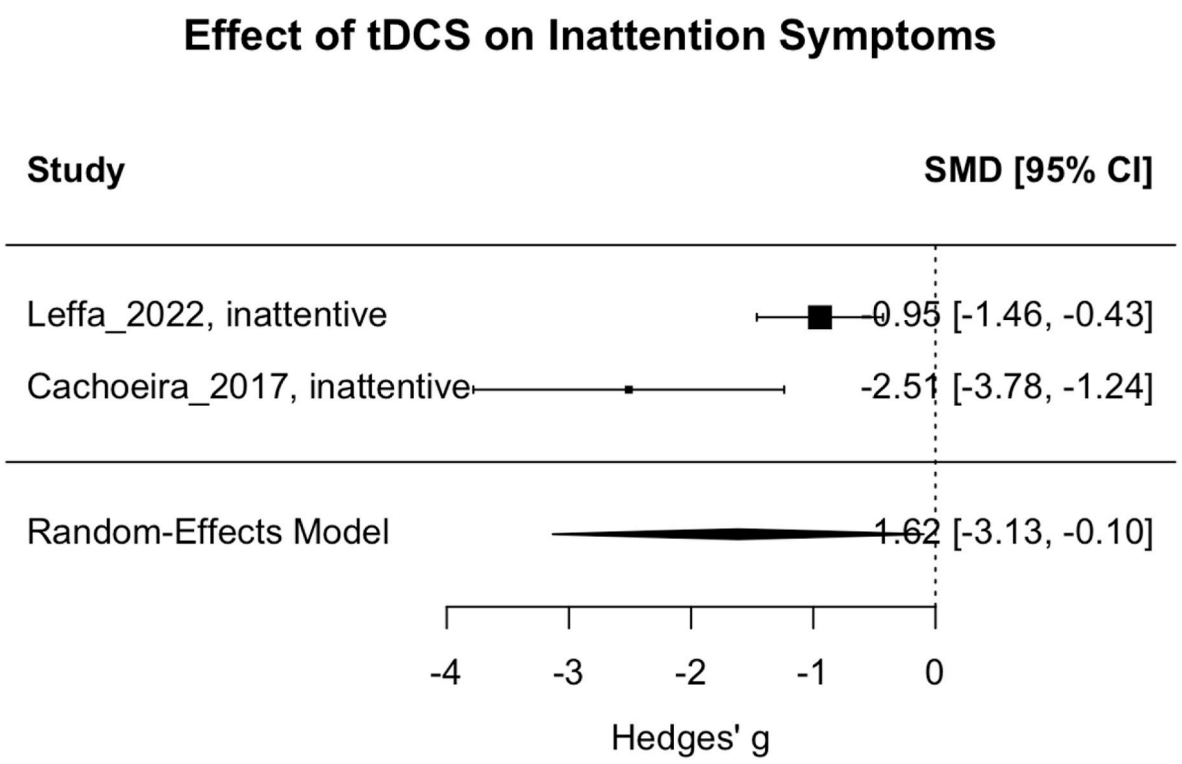


Fig. 2. Forest plot of the effect of tDCS on inattention symptoms compared to sham stimulation.

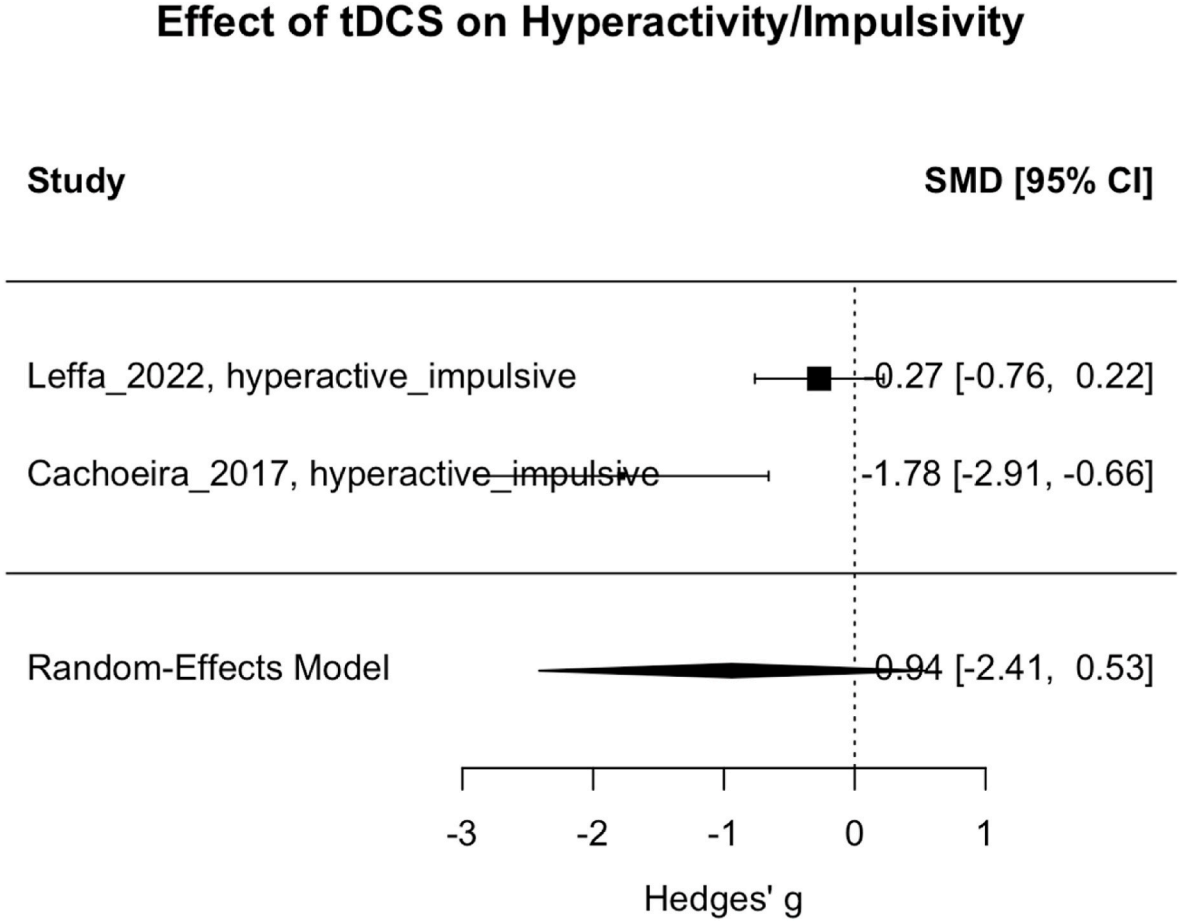


Fig. 3. Forest plot of the effect of tDCS on hyperactivity/impulsivity symptoms compared to sham stimulation.

Downloaded for Anonymous User (n/a) at Stony Brook University from ClinicalKey.com by Elsevier on October 22, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

RCT ($N = 80$) compared Dialectical Behavior Therapy (DBT) alone, tDCS alone, combined DBT and tDCS, and a control condition, employing a stimulation protocol that differed in both dosage and electrode polarity from other studies. The tDCS group received 1.5 mA stimulation (20 min/session, 10 sessions with 72-h intervals) targeting F3 (anodal) and F4 (cathodal), a reversal from other studies. Results indicated tDCS alone significantly improved inattention symptoms ($\eta^2 = .467, p = .001$) but not hyperactivity/impulsivity. Conversely, DBT showed stronger effects on hyperactivity, while the combined group demonstrated moderate improvements in both domains ($\eta^2 = .448$ for hyperactivity, .062 for inattention). While tDCS alone was also associated with significant improvements in inattention symptoms, the study's design and reporting structure precluded inclusion in the pooled analysis, highlighting substantial methodological diversity across the clinical evidence base.

Collectively, the available evidence suggests that tDCS may exert differential effects across symptom domains, with more consistent signals observed for inattention than hyperactivity/impulsivity symptoms. However, the extremely small number of studies, substantial methodological heterogeneity, wide confidence intervals, and inability to assess publication bias severely limit the interpretability of pooled estimates. Accordingly, these exploratory findings should be viewed as hypothesis-generating rather than confirmatory, highlighting the need for adequately powered trials with standardized outcome measures and longer follow-up periods.

3.3. Cognitive performance outcomes

We synthesized cognitive effects by categorizing outcomes into five domains based on the primary cognitive process targeted by each task, as defined in established neurocognitive models and task validation literature: inhibitory control, working memory, attention, executive function, and social/emotional cognition. This domain-based classification was intended to accommodate heterogeneity in task design while enabling structured comparison across studies.

Tasks were assigned to domains according to their dominant cognitive demands rather than their surface characteristics or outcome labels. For example, the Trail Making Test Part A (TMT-A) was classified under attention due to its emphasis on sustained attention, visual scanning, and processing speed, whereas the Trail Making Test Part B (TMT-B) and the Tower of London (ToL) were classified under executive function given their reliance on cognitive flexibility, set-shifting, and planning processes.

Therefore, inhibitory control was measured using the Stop Signal Task (SST), Go/No-Go Task (GNG), Stroop Task, and Emotional Flanker Task (EFT), evaluating the capacity to suppress automatic or prepotent responses —a core impairment of ADHD. Working memory was assessed using N-back tasks and Digit Span (forward and backward), with a focus on temporary storage and information manipulation. Attention was measured using Continuous Performance Test (CPT), CPT variants (Conners' CPT, MOXO-CPT), and Trail Making Test Part A (TMT-A), targeting sustained attention and vigilance. Executive function was

Table 2
Summary of cognitive task outcomes across studies.

Study	Arms (n)	Inhibitory Control	Working Memory	Attention	Executive Function	Social-Emotion	Follow-up
Basiri and Hadianfard (2023)	DBT (n = 20); tDCS (n = 20); DBT + tDCS (n = 20); Control (n = 20)	Stroop: DBT + tDCS, DBT and tDCS (all Sig vs. control; tDCS p = .001, DBT p = .002, DBT + tDCS p = .001)	NA	CPT: DBT + tDCS, DBT and tDCS (all Sig vs. control; all p = .001)	NA	NA	No follow-up beyond post-treatment
Allenby et al. (2018)	Active tDCS (n = 37); Sham (n = 37); crossover design	SST: No stim × session interaction (ns, p > .05)	N-back: No stim × session interaction (ns, p > .05)	Conners' CPT False Positive Errors (Sig, p < .001); other CPT metrics (ns, p > .05)	NA	NA	No sustained effect at follow-up (p > .05)
Barham et al. (2022)	Active tDCS (n = 11); Sham (n = 11)	Stroop IT (ns, p = .158); FR (ns, p = .915); SC (ns, p = .15)	DSB (Sig, p = .017); DSF (ns, p = .408)	CPT-CR (ns, p = .547); CPT-CRL (ns, p = .317); CPT-CE (ns, p = .346); CPT-OE (ns, p = .343); TMT-A (ns, p = .147)	ToL-TMS (Sig, p = .017); ToL-TCS (Sig, p = .033); ToL-TIT (ns, p = .083); ToL-TET (ns, p = .053); ToL-TPST (ns, p = .096); TMT-B (ns, p = .117)	RMET (ns, p = .498)	No follow-up beyond post-treatment
Beiman et al. (2025)	Final sample sizes differed by task (CPT n = 51; N-back n = 53; Stroop n = 53)	SST invalid performance; Stroop facilitation/interference (ns, all p ≥ .30)	N-back OR (Sig, p < .01); N-back CR (ns, p = .31)	CPT-OR (Sig, p = .01); CPT-CR (ns, p = .69)	NA	NA	Single session, no follow-up
Cosmo et al. (2015)	Active tDCS (n = 30); Sham (n = 30)	GNG Impulsivity errors: Fruits (ns, p = .98); Letters (ns, p = .68)	NA	GNG Omission errors: Fruits (ns, p = .89); Letters (ns, p = .79)	NA	NA	Single session, no follow-up
Dubreuil-Vall et al. (2021)	Active tDCS (n = 40); Sham (n = 40); crossover design	SST-SSRT (ns, p = .659); EFT incongruent RT (Sig, p < .001); EFT congruent RT (ns, p > .05)	NA	NA	NA	NA	Single session, no follow-up
Weinberg et al. (2023)	Active tDCS (n = 11); Sham (n = 11)	SST-SSRT (Sig, p = .03)	NA	NA	NA	NA	Single session, no follow-up
Jacoby and Lavidor. (2018)	Active tDCS (n = 20); Sham (n = 20); crossover design	MOXO-CPT Hyperactivity (Sig, p = .005); Impulsivity (ns, p = .31)	NA	MOXO-CPT Attention score (ns, p = .86)	NA	NA	Single session, no follow-up; significant learning effects across sessions masked most stimulation effects

Note. Sig: significant group difference (tDCS vs Control/Sham) at post-intervention (p < .05); ns: no significant group difference (p ≥ .05); NR: not explicitly reported; NA: not assessed. SSRT: stop signal reaction time; RT: reaction time; IT: interference time; FR: false response; SC: spontaneous correction; CR: Correct response; CRL: Correct response latency; CE: Commission Error; OE: Omission Error; TMS: Total Move score; TCS: Total Correct score; TIT: Total Initiation time; TET: Total Execution time; TPST: Total Planning-Solving time; OR: Omission Rate; CR: Commission Rate.

Downloaded for Anonymous User (n/a) at Stony Brook University from ClinicalKey.com by Elsevier on October 22, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

evaluated using the Trail Making Test Part B (TMT-B) and the Tower of London (ToL), which assess cognitive flexibility, set-shifting, and planning abilities. Social-emotional cognition was examined using the Reading the Mind in the Eyes Test (RMET), which captures emotion recognition and mental state attribution. Study mapping according to assessed domains and corresponding outcomes is detailed in Table 2, summarizing cognitive tasks, study arms compared, and reported significance at post-intervention. Reported significance classifications reflect outcomes as described within individual studies; due to inconsistent reporting of effect sizes across studies, these indicators are presented descriptively rather than as comparative measures of effect magnitude.

To further examine patterns across studies, we aggregated task-level results by domain (Table 3), reporting the number of task applications, the proportion of tasks showing benefit, and the contributing studies and their corresponding effect sizes. Across 23 cognitive task applications from eight studies, 10 tasks (43.5 %) demonstrated positive effects post-tDCS. The overall pattern revealed modest benefits in selected domains. Effect size information in Table 3 is reported where available to characterize individual study findings, but was not used for cross-study quantitative comparison given heterogeneity in task design, outcome metrics, and reporting practices, and the limited control of practice or fatigue effects across studies.

Inhibitory control was most frequently assessed (10 tasks), with four tasks showing significant improvements and generally small to moderate effect sizes. Working memory was examined through three tasks, two of which demonstrated positive effects with small to moderate effect sizes. Attention was assessed in seven tasks, with three showing significant benefit. Executive function outcomes were mixed, with only one of two tasks reporting significant improvement. No beneficial effects were found for the single social-emotional task employed. When single tasks produced multiple outcome indices within the same domain (e.g., the Tower of London yielded five measures or multiple Continuous Performance Test indices), these were treated as a single task in both the total count and positive effect tally for the purposes of domain-level synthesis. This approach was adopted to avoid outcome inflation and to provide conservative estimates of domain-level effects while preserving consistency across studies.

Table 3
Domain-level synthesis of cognitive task outcomes.

Cognitive Domain	Tasks Used	Number of Tasks	Tasks Reporting Benefit	Effect Sizes Reported
Inhibitory Control	SST, GNG, Stroop, EFT	10	4/10 positive	SST: $\eta^2 = .12$ (Weinberg et al., 2023); Stroop: $\eta^2 = .274$ (Basiri et al., 2023)
Working Memory	N-back, Digit Span (DSF, DSB)	3	2/3 positive	DSB: $\eta^2 = .211$ (Barham et al., 2022); N-back OR: $d = 1.15$ (Beiman et al., 2025)
Attention	CPT, Conners' CPT, MOXO-CPT, TMT-A	7	3/7 positive	CPT: $\eta^2 = .471$ (Basiri et al., 2023); Conners' CPT False Pos: $\beta = -.36$ (Allenby et al., 2018); CPT OR: $d = 1.11$ (Beiman et al., 2025)
Executive Function	TMT-B, ToL	2	1/2 positive	ToL Initiation Time: $\eta^2 = .434$, ToL Total Move Score: $\eta^2 = .589$ (Barham et al., 2022)
Social-Emotion	RMET	1	0/1 positive	None reported

Note. η^2 = partial eta squared; d = Cohen's d; β = standardized regression coefficient. Different effect size metrics were reported across studies depending on statistical analysis employed.

3.4. Neurophysiological outcomes

Two studies investigated neurophysiological outcomes using EEG-based measures to assess changes in brain function following tDCS.

Cosmo et al. (2015b) examined cortical connectivity using a Functional Cortical Network (FCN) model, reporting increased EEG-based functional connectivity following single-session 1 mA anodal stimulation over left DLPFC. Connectivity enhancements extended beyond the stimulated frontal site to occipital, bilateral temporal, and centro-parietal regions, suggesting diffuse modulatory effects across large-scale brain networks. Dubreuil-Vall et al. (2021) analysed event-related potentials (ERPs) during cognitive control tasks, finding that 2 mA left DLPFC stimulation increased P300 amplitude and reduced reaction time in the Eriksen Flanker Task, indicating improved conflict monitoring and attentional control. However, no significant effects were observed on ERP markers during the Stop Signal Task. The authors noted potential state-dependent effects, as baseline P300 amplitude appeared to predict treatment response.

These findings provide preliminary, mechanistic indications that prefrontal tDCS may influence neural markers of attention and cognitive control in adults with ADHD. However, given limited studies and methodological heterogeneity, neurophysiological mechanisms underlying tDCS effects remain inconclusive and warrant further investigation.

3.5. Medication status and moderators of effect

Among included studies, medication status varied considerably: 3 recruited only unmedicated participants (medication-free or ≥1 month washout), 4 recruited participants withdrawing medication on stimulation days, 1 included only medication-naïve individuals, 1 maintained stable medication status, 1 involved partially medicated participants without medication control, and 1 did not report medication details. Medication categories are reported here according to the predefined definitions outlined in Table 1. Given the small number of studies and substantial heterogeneity in medication definitions and study designs, no formal moderator analyses were performed.

Only Allenby et al. (2018) systematically explored differential tDCS effects between medicated and unmedicated participants. They found CPT false positive errors significantly improved following tDCS only among participants currently using ADHD medication ($\chi^2 = 12.15$; $p < .001$), with no comparable effect in the unmedicated group. Medication status significantly predicted CPT performance across conditions ($\beta = -.69$; $p = .001$), though three-way interaction testing was limited by sample size. These findings suggest concurrent psychostimulant use may modulate tDCS-related behavioural effects on attentional control; however, this observation is based on a single study and requires confirmation in adequately powered samples.

Additional potential moderators include age, sex, psychiatric comorbidities, and ADHD subtype. Participant age ranged widely (mean 22–38 years), with male proportions varying substantially (21.3 %–70 %). All studies excluded psychiatric comorbidities. Regarding ADHD subtypes, three studies included combined and inattentive subtypes, two included all three subtypes, and six did not specify subtype information. Despite sample composition differences, no studies conducted subgroup analyses based on demographic or clinical characteristics. The potential moderating role of medication status and other sample characteristics on tDCS efficacy remains unclear and represents a key priority for future, adequately powered studies.

3.6. Adverse events and tolerability

Adverse events were reported in 7 of 11 studies. Four studies (Barham et al., 2022; Cosmo et al., 2015a; Cosmo et al., 2015b;

Downloaded for Anonymous User (n/a) at Stony Brook University from ClinicalKey.com by Elsevier on October 22, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

Dubreuil-Vall et al., 2021; Jacoby et al., 2018) reported no adverse effects. Among them, only Cosmo et al. (2015a; 2015b) used a formal tDCS Adverse Events Questionnaire to confirm the absence of discomfort, while the remaining two studies (Dubreuil-Vall et al., 2021; Jacoby et al., 2018) did not specify how adverse effects were assessed.

Three studies provided more detailed evaluations. Leffa et al. (2022) and Cachoeira et al. (2017) systematically monitored adverse effects using the Comfort Rating Questionnaire and a tDCS Adverse Events Questionnaire, respectively. Both found mild and transient effects, such as tingling, itching, burning, and scalp discomfort, with higher frequency in the active stimulation group. No participants withdrew in Leffa et al. but in Cachoeira et al. one participant discontinued after experiencing acute mood changes post-stimulation. Allenby et al. (2018) also administered the tDCS Effects Questionnaire and reported only mild side effects. Additionally, Beiman et al. (2025) conducted follow-up phone assessments at 5 and 24 h post-session, with no major concerns identified.

Overall, the evidence indicates that tDCS is generally safe and well-tolerated in adults with ADHD. However, adverse event assessment methods were not standardized across studies, limiting comparability of safety profiles and constraining interpretation of tolerability outcomes, which underscore the need for more systematic and longitudinal assessment in future studies. Future studies would benefit from the use of standardized adverse event reporting tools to improve consistency across trials.

3.7. Stimulation parameters

Included studies employed heterogeneous tDCS protocols varying across electrode montage, intensity (1–2 mA), session duration (10–30 min), total sessions (single session to 28 sessions), and stimulation schedule (daily vs. weekly). The most common approach was anodal DLPFC stimulation, with five studies targeting left DLPFC (F3), three targeting right DLPFC (F4), one applying bilateral anodal stimulation over F3 and F4, and one using an anode over the F4-F8 crossing point. Details are provided in Table 1. Substantial variability in montage, polarity, and session structure precludes firm conclusions regarding optimal stimulation parameters. No clear pattern emerged linking specific parameter combinations with treatment efficacy.

Only Dubreuil-Vall et al. (2021) employed crossover design to directly compare left and right DLPFC stimulation, reporting that right-sided stimulation (F4) led to significantly greater commission error reductions, suggesting potential lateralized effects on attention-related outcomes. However, these findings are preliminary and have not been replicated.

No studies used individualized protocols or adapted parameters based on participant-specific neurocognitive or neurophysiological profiles. Given methodological heterogeneity and limited sample sizes, current evidence does not support clear conclusions regarding optimal stimulation parameters. Further trials are needed to systematically evaluate electrode placement, polarity, intensity, and potential personalized tDCS approaches in adult ADHD populations.

4. Discussion

This systematic review synthesized current evidence regarding the efficacy and tolerability of transcranial direct current stimulation (tDCS) in adults with Attention-Deficit/Hyperactivity Disorder (ADHD). Across the eleven included studies, considerable methodological heterogeneity was observed, limiting the strength of inferences that can be drawn. Nevertheless, patterns within the available evidence allow tentative observations regarding the domains in which effects have been examined, while acknowledging that the strength and consistency of these effects remain uncertain.

4.1. Clinical efficacy: differential effects on core symptoms

Findings from the exploratory meta-analysis suggested a differential response pattern between ADHD symptom domains. While pooled estimates indicated a potential reduction in inattention symptoms following tDCS, no statistically significant benefit was observed for hyperactivity/impulsivity. Given the limited number of studies contributing to the quantitative synthesis and the substantial heterogeneity observed, these findings should be interpreted cautiously. Nonetheless, the directional pattern is broadly consistent with reports from pediatric populations (Salehinejad et al., 2020), suggesting that observed effects may vary across symptom domains rather than indicating uniform clinical benefit.

This apparent differential pattern may reflect differences in the neurobiological substrates underlying ADHD symptom domains. Inattention has been closely linked to dysfunction within prefrontal cortical networks involved in executive control, the brain regions that are relatively accessible to superficial cortical stimulation techniques such as tDCS. In contrast, hyperactivity and impulsivity are thought to involve more distributed cortico-subcortical circuits, including deep structures such as the basal ganglia, which are less directly targeted by conventional tDCS protocols. From a mechanistic perspective, these considerations are consistent with the possibility that tDCS effects in adult ADHD may be more readily detectable in cognitive-attentional domains than in behavioural symptom domains, although this interpretation remains tentative given current evidence limitations.

4.2. Cognitive outcomes: task-specific improvements

Cognitive assessments revealed a nuanced pattern of improvements that extended beyond simple domain-level analyses. Working memory and inhibitory control tasks, particularly the N-back and Stop Signal Task, were associated with the most consistently reported task-level improvements following tDCS. However, effects were often confined to specific task indices rather than global performance improvements. For instance, significant reductions in omission errors occurred without corresponding changes in commission errors on both the N-back and CPT tasks (Beiman et al., 2025), highlighting the precision with which tDCS may influence distinct cognitive processes.

This granular pattern of improvement underscores the importance of moving beyond broad cognitive domains toward more refined outcome measures. Rather than aggregating effects at the task level, future studies should identify which specific cognitive processes, sustained attention versus inhibitory speed, for example, are most responsive to prefrontal stimulation. Such precision would not only enhance mechanistic understanding but also facilitate the development of personalized treatment protocols tailored to individual cognitive profiles.

The relatively modest effects observed in adult populations, compared to more pronounced benefits reported in studies including pediatric populations (Salehinejad et al., 2019), may reflect developmental differences in neural plasticity and cortical network organization. These findings suggest that therapeutic expectations and protocols may need to be adjusted for adult populations, with greater emphasis on sustained treatment regimens or alternative stimulation approaches.

4.3. Neurophysiological insights: an underexplored frontier

Only two included studies incorporated neurophysiological assessments, both using EEG and reported alterations in neural oscillatory activity following tDCS. Specifically, observed changes in spectral power and event-related potential components (e.g., P200/P300) are consistent with modulation of attentional allocation, stimulus evaluation, and cognitive control processes, which align with the cognitive domains assessed behaviourally in this review. Similarly, reported changes in functional connectivity and frontal oscillatory patterns may reflect engagement of prefrontal control networks implicated in executive functioning and sustained attention.

Downloaded for Anonymous User (n/a) at Stony Brook University from ClinicalKey.com by Elsevier on October 22, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

However, both EEG studies employed single-session stimulation designs, limiting the extent to which these findings can be interpreted as markers of sustained neurophysiological change or long-term clinical benefit. As such, while these results provide preliminary mechanistic indications of tDCS-related modulation of neural processes relevant to ADHD, conclusions regarding enduring neuroplastic effects remain constrained.

The limited integration of objective neurobiological markers in adult tDCS research therefore represents a significant gap in the literature. In contrast, pediatric tDCS research has increasingly integrated EEG and neuroimaging modalities to link neural changes with cognitive and clinical outcomes (Westwood et al., 2021b), enabling more sophisticated mechanistic insights and personalized treatment approaches.

Future adult studies should prioritize the inclusion of neurophysiological assessments, such as event-related potentials (ERPs), resting-state EEG, and functional near-infrared spectroscopy (fNIRS), which are feasible, cost-effective, and well-tolerated in adult populations, within multi-session and longitudinal designs. These tools can serve dual purposes: providing mechanistic insights into how tDCS modulates neural activity, and identifying biomarkers of treatment response to support stratified or adaptive intervention models. For instance, EEG-based indices of frontal theta activity or connectivity patterns may help identify individuals most likely to benefit from prefrontal stimulation, while longitudinal changes in these signals could offer early indicators of treatment efficacy. These approaches would allow examination of whether acute neural changes (e.g., P300 amplitude modulation or connectivity shifts) translate into sustained cognitive improvement or clinical symptom reduction over time. Embedding neurophysiological monitoring into tDCS protocols may therefore support both mechanistic understanding and the development of stratified or adaptive intervention models in adult ADHD.

4.4. Stimulation parameters: the challenge of optimization

The substantial variability in stimulation protocols across studies, including electrode placement, current intensity, session duration, and treatment frequency, precluded clear conclusions about optimal parameters. While most studies targeted the left dorsolateral prefrontal cortex (DLPFC, F3) with anodal stimulation, systematic comparisons of montages, intensities, and dosing schedules remain limited. Only one study (Dubreuil-Vall et al., 2021) used a within-subject crossover design to compare left and right DLPFC stimulation, reporting no statistically significant differences in outcomes.

The question of treatment durability emerged as particularly important. The single study (Allenby et al., 2018) including follow-up assessment found that cognitive benefits did not persist after stimulation discontinuation, suggesting that sustained or booster protocols may be necessary for long-term efficacy. This finding has important implications for clinical implementation, as it suggests that tDCS for adult ADHD may require ongoing maintenance therapy rather than discrete treatment courses.

4.5. Clinical implications and future directions

These findings have several important clinical implications. First, tDCS may be more appropriately considered as a potential adjunctive approach targeting specific cognitive-attentional processes, rather than a comprehensive ADHD intervention. Second, the need for sustained treatment protocols has important cost-effectiveness considerations that warrant systematic evaluation. Third, the safety profile reported across studies suggests that extended treatment regimens may be feasible, although standardized adverse event reporting remains necessary.

Several critical research priorities emerge from this review. Large-scale, adequately powered trials with harmonized protocols are urgently needed to establish definitive efficacy and identify optimal stimulation parameters. Long-term follow-up studies should evaluate treatment durability and establish guidelines for maintenance therapy. Critically, future studies must incorporate stratified analyses by ADHD subtype, sex, age, and medication status to identify individual difference factors that moderate treatment response.

Moreover, head-to-head comparisons with established non-pharmacological interventions, such as cognitive behavioural therapy or neurofeedback, would help position tDCS within the broader treatment landscape for adult ADHD. Economic evaluations comparing tDCS to standard care approaches would inform implementation decisions and healthcare policy.

4.6. Limitations

Several limitations constrain the interpretation of these findings. The small number of included studies and substantial methodological heterogeneity limited our ability to conduct robust quantitative syntheses, where quantitative pooling was possible, it was restricted to an exploratory meta-analysis based on two trials. Most studies employed small sample sizes and lacked long-term follow-up, which restricted generalizability and precluded evaluation of treatment durability. The wide variability in outcome measures, coupled with limited consideration of potential moderator variables, further constrains the strength of conclusions. Methodological quality was variable and generally moderate to high, although statistical power was frequently limited, reflecting small sample sizes across trials.

Additionally, the risk of publication bias and selective reporting cannot be excluded, particularly given the early stage of this research field and the predominance of pilot studies, which often involve narrative reporting of results.

CRediT authorship contribution statement

Tianran Zhang: Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Dimitrios Adamis:** Writing – review & editing, Supervision, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that there are no conflicts of interest regarding the publication of this article.

Acknowledgements

All authors listed have made a substantial, direct, and intellectual contribution to the work and approved it for publication. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jpsychires.2026.01.030.

References

Adamis, D., Flynn, C., Wrigley, M., Gavin, B., McNicholas, F., 2022. ADHD in adults: a systematic review and meta-analysis of prevalence studies in outpatient psychiatric clinics. J. Atten. Disord. 26 (12), 1523–1534. https://doi.org/10.1177/10870547221085503.

Allenby, C., Falcone, M., Bernardo, L., Wileyto, E.P., Rostain, A., Ramsay, J., Lerman, C., Loughead, J., 2018. Transcranial direct current brain stimulation decreases impulsivity in ADHD. Brain Stimul. 11 (5), 974–981. https://doi.org/10.1016/j.brs.2018.04.016.

American Psychiatric Association, 2013. Diagnostic and Statistical Manual of Mental Disorders. American psychiatric association.

Barham, H., Büyükgök, D., Aksu, S., Soyata, A.Z., Bulut, G., Eskicioğlu, G., Kulaksızoğlu, I.B., 2022. Evidence for modulation of planning and working memory capacities by transcranial direct current stimulation in a sample of adults with

Downloaded for Anonymous User (n/a) at Stony Brook University from ClinicalKey.com by Elsevier on October 22, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.

attention deficit hyperactivity disorder. Neurosci. Lett. 790, 136883. https://doi.org/10.1016/j.neulet.2022.136883.
Barkley, R.A., 1997. Behavioral inhibition, sustained attention, and executive functions: constructing a unifying theory of ADHD. Psychol. Bull. 121 (1), 65–94. https://doi.org/10.1037/0033-2909.121.1.65.
Basiri, N., Hadianfard, H., 2023. Adult ADHD treatment based on combination of dialectical behavior therapy (DBT) and transcranial direct current stimulation (tDCS) as measured by subjective and objective scales. J. Atten. Disord. 27 (1), 57–66. https://doi.org/10.1177/10870547221118527.
Beiman, N., Eitan, R., Todder, D., Keha, E., Kalanthroff, E., 2025. The effect of transcranial direct Current stimulation (tDCS) on cognitive functioning in individuals with Attention-Deficit/Hyperactivity disorder (ADHD). PubMed 22 (1), 87–98. https://doi.org/10.36131/cnfioritieditore20250107.
Bikson, M., Grossman, P., Thomas, C., Zannou, A.L., Jiang, J., Adnan, T., Mourdoukoutas, A.P., Kronberg, G., Truong, D., Boggio, P., Brunoni, A.R., Charvet, L., Fregni, F., Fritsch, B., Gillick, B., Hamilton, R.H., Hampstead, B.M., Jankord, R., Kirton, A., Woods, A.J., 2016. Safety of transcranial direct current stimulation: evidence Based Update 2016. Brain Stimul. 9 (5), 641–661. https://doi.org/10.1016/j.brs.2016.06.004.
Breitling, C., Zaehle, T., Dannhauer, M., Bonath, B., Tegelbeckers, J., Flechtner, H., Krauel, K., 2016. Improving interference control in ADHD patients with transcranial direct Current stimulation (tDCS). Front. Cell. Neurosci. 10. https://doi.org/10.3389/fncel.2016.00072.
Cachoeira, C.T., Leffa, D.T., Mittelstadt, S.D., Mendes, L.S.T., Brunoni, A.R., Pinto, J.V., Blazius, V., Machado, V., Bau, C.H.D., Rohde, L.A., Grevet, E.H., Schestatsky, P., 2017. Positive effects of transcranial direct current stimulation in adult patients with attention-deficit/hyperactivity disorder A pilot randomized controlled study. Psychiatry Res. 247, 28–32. https://doi.org/10.1016/j.psychres.2016.11.009.
Cosmo, C., Baptista, A.F., De Araújo, A.N., Rosário, R.S.D., Miranda, J.G.V., Montoya, P., De Sena, E.P., 2015. A randomized, Double-Blind, SHAM-Controlled trial of transcranial direct current stimulation in Attention-Deficit/Hyperactivity Disorder. PLoS One 10 (8), e0135371. https://doi.org/10.1371/journal.pone.0135371.
Cosmo, C., Ferreira, C., Miranda, J.G.V., Do Rosario, R.S., Baptista, A.F., Montoya, P., De Sena, E.P., 2015b. Spreading effect of tDCS in individuals with attention-deficit/hyperactivity disorder as shown by functional cortical networks: a randomized, double-blind, sham-controlled trial. Front. Psychiatr. 6, 111.
Courrèges, M., Hoareau, M., Levenes, C., Rahioui, H., 2025. Comparative efficacy of neurofeedback, tDCS, and TMS: the future of therapy for adults with ADHD. A systematic review and meta-analysis. J. Affect. Disord. 388, 119585. https://doi.org/10.1016/j.jad.2025.119585.
Daley, D., Van der Oord, S., Ferrin, M., Danckaerts, M., Doepfner, M., Cortese, S., et al., 2014. Behavioral interventions in attention-deficit/hyperactivity disorder: a meta-analysis of randomized controlled trials across multiple outcome domains. J. Am. Acad. Child Adolesc. Psychiatr. 53 (8), 835–847.
Downs, S.H., Black, N., 1998. The feasibility of creating a checklist for the assessment of the methodological quality both of randomised and non-randomised studies of health care interventions. J. Epidemiol. Community 52 (6), 377–384. https://doi.org/10.1136/jech.52.6.377.
Dubreuil-Vall, L., Gomez-Bernal, F., Villegas, A.C., Cirillo, P., Surman, C., Ruffini, G., Widge, A.S., Camprodon, J.A., 2021. Transcranial direct current stimulation to the left dorsolateral prefrontal cortex improves cognitive control in patients with Attention-Deficit/Hyperactivity Disorder: a randomized behavioral and neurophysiological study. Biol. Psychiatry Cogn. Neurosci. Neuroimaging 6 (4), 439–448. https://doi.org/10.1016/j.bpsc.2020.11.006.
Fayyad, J., Sampson, N.A., Hwang, I., Adamowski, T., Aguilar-Gaxiola, S., Al-Hamzawi, A., Andrade, L.H.S.G., Borges, G., De Girolamo, G., Florescu, S., Gureje, O., Haro, J.M., Hu, C., Karam, E.G., Lee, S., Navarro-Mateu, F., O'Neill, S., Pennell, B., Piazza, M., Kessler, R.C., 2016. The descriptive epidemiology of DSM-IV adult ADHD in the World Health Organization World Mental Health surveys. ADHD Attention Deficit and Hyperactivity Disorders 9 (1), 47–65. https://doi.org/10.1007/s12402-016-0208-3.
Hsu, S., Piedra, A.N., Froehlich, T.E., Brinkman, W.B., Epstein, J., 2017. 6.18 Caregiver-Reported side effects of Methylphenidate and comorbid oppositional defiant disorder diagnosis in children with Attention-Deficit/Hyperactivity disorder (ADHD). J. Am. Acad. Child Adolesc. Psychiatr. 56 (10), S282. https://doi.org/10.1016/j.jaac.2017.09.363.
Imburgio, M.J., Orr, J.M., 2018. Effects of prefrontal tDCS on executive function: methodological considerations revealed by meta-analysis. Neuropsychologia 117, 156–166. https://doi.org/10.1016/j.neuropsychologia.2018.04.022.
Jacoby, N., Lavidor, M., 2018. Null tDCS effects in a sustained attention task: the Modulating role of learning. Front. Psychol. 9. https://doi.org/10.3389/fpsyg.2018.00476.
Kasper, L.J., Alderson, R.M., Hudec, K.L., 2012. Moderators of working memory deficits in children with attention-deficit/hyperactivity disorder (ADHD): a meta-analytic review. Clin. Psychol. Rev. 32 (7), 605–617. https://doi.org/10.1016/j.cpr.2012.07.001.
Keune, P.M., Wiedemann, E., Schneidt, A., Schönenberg, M., 2015. Frontal brain asymmetry in adult attention-deficit/hyperactivity disorder (ADHD): extending the motivational dysfunction hypothesis. Clin. Neurophysiol. 126 (4), 711–720. https://doi.org/10.1016/j.clinph.2014.07.008.
Kooij, S.J., Bejerot, S., Blackwell, A., Caci, H., Casas-Brugué, M., Carpentier, P.J., Edvinsson, D., Fayyad, J., Foeken, K., Fitzgerald, M., Gaillac, V., Ginsberg, Y., Henry, C., Krause, J., Lensing, M.B., Manor, I., Niederhofer, H., Nunes-Filipe, C., Ohlmeier, M.D., Asherson, P., 2010. European consensus statement on diagnosis and treatment of adult ADHD: the European Network Adult ADHD. BMC Psychiatry 10 (1). https://doi.org/10.1186/1471-244x-10-67.
Krain, A.L., Castellanos, F.X., 2006. Brain development and ADHD. Clin. Psychol. Rev. 26 (4), 433–444. https://doi.org/10.1016/j.cpr.2006.01.005.
Leffa, D.T., Grevet, E.H., Bau, C.H.D., Schneider, M., Ferrazza, C.P., Da Silva, R.F., Miranda, M.S., Picon, F., Teche, S.P., Sanches, P., Pereira, D., Rubia, K., Brunoni, A.R., Camprodon, J.A., Caumo, W., Rohde, L.A., 2022. Transcranial direct Current stimulation vs sham for the treatment of inattention in adults with Attention-Deficit/Hyperactivity disorder. JAMA Psychiatry 79 (9), 847. https://doi.org/10.1001/jamapsychiatry.2022.2055.
Marcos-Vidal, L., Martínez-García, M., Pretus, C., Garcia-Garcia, D., Martínez, K., Janssen, J., Vilarroya, O., Castellanos, F.X., Desco, M., Sepulcre, J., Carmona, S., 2018. Local functional connectivity suggests functional immaturity in children with attention-deficit/hyperactivity disorder. Hum. Brain Mapp. 39 (6), 2442–2454. https://doi.org/10.1002/hbm.24013.
Page, M.J., McKenzie, J.E., Bossuyt, P.M., Boutron, I., Hoffmann, T.C., Mulrow, C.D., Shamseer, L., Tetzlaff, J.M., Akl, E.A., Brennan, S.E., Chou, R., Glanville, J., Grimshaw, J.M., Hróbjartsson, A., Lalu, M.M., Li, T., Loder, E.W., Mayo-Wilson, E., McDonald, S., Moher, D., 2021. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. Br. Med. J. n71. https://doi.org/10.1136/bmj.n71.
Passarotti, A.M., Sweeney, J.A., Pavuluri, M.N., 2009. Neural correlates of response inhibition in pediatric bipolar disorder and attention deficit hyperactivity disorder. Psychiatry Res. Neuroimaging. 181 (1), 36–43. https://doi.org/10.1016/j.pscychresns.2009.07.002.
Polanía, R., Nitsche, M.A., Ruff, C.C., 2018. Studying and modifying brain function with non-invasive brain stimulation. Nat. Neurosci. 21 (2), 174–187. https://doi.org/10.1038/s41593-017-0054-4.
Rapport, M.D., Orban, S.A., Kofler, M.J., Friedman, L.M., 2013. Do programs designed to train working memory, other executive functions, and attention benefit children with ADHD? A meta-analytic review of cognitive, academic, and behavioral outcomes. Clin. Psychol. Rev. 33 (8), 1237–1252. https://doi.org/10.1016/j.cpr.2013.08.005.
Salehinejad, M.A., Wischnewski, M., Nejati, V., Vicario, C.M., Nitsche, M.A., 2019. Transcranial direct current stimulation in attention-deficit hyperactivity disorder: a meta-analysis of neuropsychological deficits. PLoS One 14 (4), e0215095. https://doi.org/10.1371/journal.pone.0215095.
Salehinejad, M.A., Nejati, V., Mosayebi-Samani, M., Mohammadi, A., Wischnewski, M., Kuo, M., Avenanti, A., Vicario, C.M., Nitsche, M.A., 2020. Transcranial Direct Current Stimulation in ADHD: a systematic review of efficacy, safety, and protocol-induced electrical field modeling results. Neurosci. Bull. 36 (10), 1191–1212. https://doi.org/10.1007/s12264-020-00501-x.
Samea, F., Soluki, S., Nejati, V., Zarei, M., Cortese, S., Eickhoff, S.B., Tahmasian, M., Eickhoff, C.R., 2019. Brain alterations in children/adolescents with ADHD revisited: a neuroimaging meta-analysis of 96 structural and functional studies. Neurosci. Biobehav. Rev. 100, 1–8. https://doi.org/10.1016/j.neubiorev.2019.02.011.
Seidman, L.J., Valera, E.M., Makris, N., 2005. Structural brain imaging of Attention-Deficit/Hyperactivity disorder. Biol. Psychiatry 57 (11), 1263–1272. https://doi.org/10.1016/j.biopsych.2004.11.019.
Sibley, M.H., Kuriyan, A.B., Evans, S.W., Waxmonsky, J.G., Smith, B.H., 2014. Pharmacological and psychosocial treatments for adolescents with ADHD: an updated systematic review of the literature. Clin. Psychol. Rev. 34 (3), 218–232. https://doi.org/10.1016/j.cpr.2014.02.001.
Sibley, M.H., Rohde, L.A., Swanson, J.M., Hechtman, L.T., Molina, B.S., Mitchell, J.T., Arnold, L.E., Caye, A., Kennedy, T.M., Roy, A., Stehli, A., 2017. Late-Onset ADHD reconsidered with comprehensive repeated assessments between ages 10 and 25. Am. J. Psychiatr. 175 (2), 140–149. https://doi.org/10.1176/appi.ajp.2017.17030298.
Turgay, A., Goodman, D.W., Asherson, P., Lasser, R.A., Babcock, T.F., Pucci, M.L., Barkley, R., 2012. Lifespan persistence of ADHD. J. Clin. Psychiatry 73 (2), 192–201. https://doi.org/10.4088/jcp.10m06628.
Weinberg, H., Baruch, Y., Tzameret, H., Lavidor, M., 2023. Cognitive control enhancement in attention deficit hyperactivity disorder (ADHD) and neurotypical individuals. Exp. Brain Res. 241 (9), 2381–2392. https://doi.org/10.1007/s00221-023-06695-6.
Westwood, S.J., Radua, J., Rubia, K., 2021. Noninvasive brain stimulation in children and adults with attention-deficit/hyperactivity disorder: a systematic review and meta-analysis. J. Psychiatr. Neurosci. 46 (1), E14–E33. https://doi.org/10.1503/jpn.190179.
Westwood, S.J., Bozhilova, N., Criaud, M., Lam, S., Lukito, S., Wallace-Hanlon, S., Kowalczyk, O.S., Kostara, A., Mathew, J., Wexler, B.E., Kadosh, R.C., Asherson, P., Rubia, K., 2021b. The effect of transcranial direct current stimulation (tDCS) combined with cognitive training on EEG spectral power in adolescent boys with ADHD: a double-blind, randomized, sham-controlled trial. IBRO Neuroscience Reports 12, 55–64. https://doi.org/10.1016/j.ibneur.2021.12.005.
Willcutt, E.G., Doyle, A.E., Nigg, J.T., Faraone, S.V., Pennington, B.F., 2005. Validity of the executive function theory of Attention-Deficit/Hyperactivity disorder: a meta-analytic review. Biol. Psychiatry 57 (11), 1336–1346. https://doi.org/10.1016/j.biopsych.2005.02.006.

Downloaded for Anonymous User (n/a) at Stony Brook University from ClinicalKey.com by Elsevier on October 22, 2026. For personal use only. No other uses without permission. Copyright ©2026. Elsevier Inc. All rights reserved.